

KARTIK DESHPANDE

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EXPERIENCE

Google Summer of Code – VCell (NRNB)

May 2026 – Present

Open Source Contributor

Remote

- Integrating **Auth0** with cross-platform identity sharing between VCell and VCell-AI, enabling unified accounts, shared token tracking, and seamless login across both platforms.
- Implementing per-user token limits with an **admin dashboard**, refactoring the prompt engineering layer, and extending local LLM support via **Ollama** with a model selector UI and dynamic embedding dimension detection.
- Adding **BioModels DB** and **PubMed** search integrations, expanding support to SBML/BioPAX data queries, and building model design assistance for generating and combining biological models.

Motion Education

Jun 2025 – Oct 2025

Software Development Intern (Full Stack)

Remote

- Developed and deployed a scalable **FastAPI backend** for an AI-powered academic assistant, integrating **MongoDB**, **Pinecone vector DB**, and **OpenAI GPT models** to improve query accuracy.
- Engineered **ingestion pipelines** to transform NCERT textbooks, PDF modules, and video lectures into embeddings for efficient **retrieval-augmented generation (RAG)**.
- Implemented **context-aware session management** with **chat memory**, **summarization**, and **vector search**, enhancing response relevance and personalization.

Dukaan

Aug 2024 – Oct 2024

Software Development Intern (Full Stack)

Bangalore

- Enhanced the BOT9 chat module using the **MERN** stack; integrated **Slack webhooks**, built an OpenAI-powered chatbot with **HTTP streaming**, and designed an **analytics dashboard** with interactive graphs.

PROJECTS

Canteeno.in

- Developed and deployed a full-stack food ordering platform with **650+ registered users**, **40+ DAUs**, and **INR 2L+** revenue from **1,500+** processed orders.
- Integrated **Razorpay payment gateway** with **webhook-based settlement** and an **in-app wallet system**, ensuring **transaction safety** via **atomic DB operations** and **preventing race conditions**.
- Designed **analytics dashboards** with **MongoDB aggregation** pipelines, enabling **revenue tracking**, **peak-hour insights**, and **business KPIs** for canteen owners.
- Engineered a **secure multi-role authentication system** featuring **email verification**, **OTP-based password resets**, and **RBAC middleware**, supporting students, vendors, and administrators.

EDUCATION

Scaler School of Technology

Jul 2023 – Jul 2027

Undergraduate Program in Computer Science

Bangalore, India

- CGR: **8.6**

Birla Institute of Technology (BITS), Pilani

Jul 2023 – Jul 2027

B.Sc. (Hons.) in Computer Science

Pilani, India

- CGPA: **8.43**

TECHNICAL SKILLS

Languages	Java, Python, C++, JavaScript
Backend & Frameworks	Node.js, Express, FastAPI, Spring Boot, React
Databases	MongoDB, MySQL, VectorDB (Pinecone)
AI	OpenAI API, RAG
Tools & Concepts	Git, Docker, Postman, WebSockets, Webhooks, SOLID Principles, Design Patterns

ACHIEVEMENTS

- Google Summer of Code 2025:** Selected contributor for VCell (NRNB org) – AI/LLM tooling for computational biology.
- Hackathon Winner:** Scaler AI Labs Hackathon – **INR 80,000** cash prize.
- Competitive Programming:** Knight on LeetCode (max. rating **1878**).
- Academic Excellence:** Dean's Merit List Award (top **10%**) and scholarship for academic excellence.
- Teaching Assistant:** DSA and DBMS at Scaler School of Technology.